FACILITATING MANUAL SEGMENTATION OF 3D DATASETS USING CONTOUR AND INTENSITY GUIDED INTERPOLATION

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ABSTRACT

Manual segmentation of anatomical structures in 3D imaging datasets is a highly time-consuming process. This process can be sped up using interslice interpolation techniques, which require only a small subset of slices to be manually segmented. In this paper, we propose a two-step interpolation approach that utilizes a "binary weighted averaging" algorithm to interpolate contour information, and the random forest framework to perform intensity-based label classification. We present the results of experiments performed in the context of hippocampal segmentations in ex vivo MRI scans. Compared to the random walker algorithm and morphology-based interpolation, the proposed method produces more accurate segmentations and smoother 3D reconstructions.

Index Terms— 3D segmentation, contour interpolation, random forest

1. INTRODUCTION

An important task in neuroimaging research is the generation of ground truth data by manually segmenting brain structures of interest in 3D images. These segmentations typically play a crucial role in the development of algorithms in morphometry studies aimed at understanding the pathophysiology of neurological disorders. Manual segmentation involves delineating the boundaries of regions of interest on serial 2D cross-sections. This process is laborious and time-consuming, especially in high-resolution images with many slices. Furthermore, serial labelling of every slice potentially introduces step-edge artifacts in the resulting segmentation.

To address these problems, interpolation techniques have been developed to propagate labels from partially annotated datasets to the remaining unlabelled slices in the dataset, thereby allowing manual segmentation of only a small subset of slices. Some methods, known as contour-based methods, use shape information extracted from the manually traced boundaries to guide the interpolation [1]. A limitation of these methods is that in regions of the image where the object of interest can be distinguished from adjacent objects using intensity features, this information is ignored. For example, in MRI scans of the cortex, the intensity values of pixels varies significantly between grey matter and white matter. In such cases, taking advantage of this information can help improve segmentation accuracy. Other methods like the random walker algorithm [2] combine intensity and positional information during interpolation, but the positional information used is more rudimentary compared to explicitly contour-based methods.

In this work, we propose integrating intensity information with a more advanced contour-based method to achieve an improved interpolation which is better able to handle complex shape deformations while minimizing image intensity related artifacts. We evaluate our method on an *ex vivo* MRI dataset of the hippocampal region. In our experiments, we compare our approach with commonly used semi-automated segmentation techniques, specifically random walker and morphology-based interpolation [3]. Our results demonstrate that the proposed interpolation method significantly improves segmentation accuracy and generates smoother 3D reconstructions of the object of interest.

2. METHOD

The proposed interpolation technique is a two step approach based on a combination of "binary weighted averaging" for contour-based interpolation [4], and random forest for intensity-based classification [5]. Using both contour and intensity information from select slices for which manual segmentations are available, segmentations for the unlabelled slices are interpolated to reconstruct a 3D representation of the object. Both methods are used to generate probability maps in which every pixel in the unlabelled slices is assigned a probability of belonging to the object of interest. The two

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approaches are integrated by computing the voxel-wise average of the probability maps generated by each method. This average result is then thresholded to obtain the most likely segmentation for each slice.

2.1. Binary Weighted Averaging for contour-based interpolation

Univariate interpolation techniques can be used for contour-based interpolation by treating binary segmentations as sets, and implicitly representing each set by its signed distance transform [1]. The signed distance transform $f(x, S)$ of a point x in a set S is the Euclidean distance between x and the boundary of the set. This distance is positive if x is within the boundary, and negative if x is outside the boundary. One way of interpolating between two sets $A, B \subset \mathbb{R}^2$ is to simply compute the binary weighted average (BWA) of their signed distance transforms and threshold the result at zero,

$$R(t) = tA \oplus (1-t)B = \{x : f_{A,B,t}(x) \geq 0\} \quad \text{where} \quad (1)$$

$$f_{A,B,t}(x) = tf(x, A) + (1-t)f(x, B) \quad (2)$$

The parameter $t \in [0, 1]$ determines the location of the slice to be interpolated with respect to the manual segmentations. This "distance average" passes through the original sets A and B when $t = 1$ and 0 respectively. A limitation of this linear approach is that it can cause uneven interpolations. To overcome this problem, Kels et al. [4] reparameterize t such that the size of the interpolated set is linear to the new parameter $t' = g^{-1}_{A,B}(t)$, where

$$g_{A,B}(t) = \text{sign}(t)\frac{|R(t)\Delta B|}{|A\Delta B|} \quad (3)$$

and $A\Delta B = (A \setminus B) \cup (B \setminus A)$. This reparametrization of t works well when interpolating between two "simply different" sets, where the difference between the sets $(A \setminus B)$ forms a single connected component. However, when dealing with more complex anatomical structures, the difference between the segmentations being interpolated generally contains multiple connected components. In such cases, the reparameterization $g^{-1}_{A,B}(t)$ is not well-defined. To address this problem, Kels et al. propose using a piece-wise interpolation approach which breaks the weighted average of two "general sets" into several averages of simply different sets (Fig. 1). The weighted average can be computed separately for each of the connected components, C_i in the difference sets $\{A \setminus B\}$ and $\{B \setminus A\}$ as follows:

$$R_{C_i}(t) = g^{-1}_{A,B}(t)A'_i \oplus (1 - g^{-1}_{A,B}(t))B'_i \quad \text{where} \quad (4)$$

$$A'_i = \begin{cases} C_i \cup (A \cap B) & \text{if } C_i \in \{A \setminus B\} \\ A \cap B & \text{if } C_i \in \{B \setminus A\} \end{cases} \quad (5)$$

$$B'_i = \begin{cases} A \cap B & \text{if } C_i \in \{A \setminus B\} \\ C_i \cup (A \cap B) & \text{if } C_i \in \{B \setminus A\} \end{cases} \quad (6)$$

since A'_i and B'_i form simply different sets. These individual averages are merged by computing $R(t) = \bigcup_i R_{C_i}(t)$ to obtain a final weighted average. Using this piece-wise approach, segmentations for intermediate slices can be obtained by interpolating between neighboring manually segmented slices and varying the parameter t.

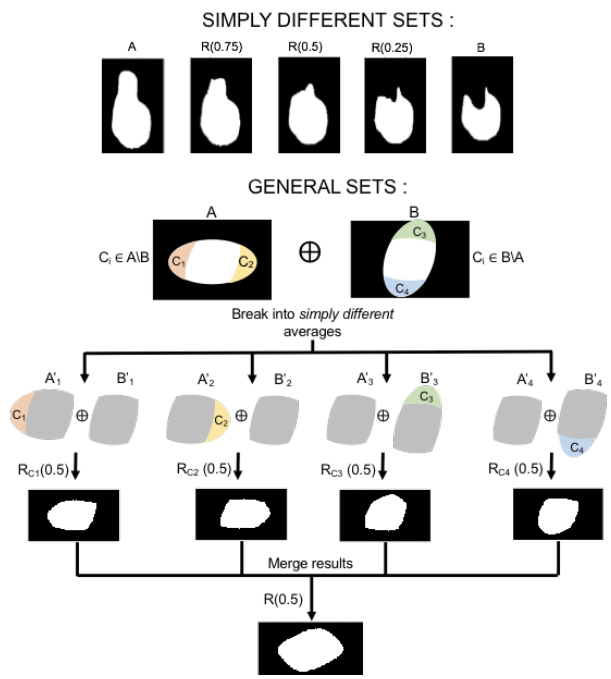

Fig. 1. Weighted average of two simply different sets for various values of the averaging parameter t, and an illustration of the approach for computing the weighted average of two general sets, when t = 0.5.

2.2. Probabilistic weighted averages

In most problems, the orientation of the slice plane on which manual segmentations are drawn relative to the orientation of the object of interest is somewhat arbitrary. Varying the direction of the slice plane, and performing repeat interpolations would provide a probabilistic interpolation that accounts for this variability in the slice plane. However, this isn't feasible since it would require performing manual segmentations anew for each transformation of the slice plane. Instead, we can approximate the transformations that the segmentations would undergo due to modification of the slicing plane by using affine transformations, or to make it even simpler, translations.

To this end, we propose performing multiple interpolations per pair of consecutive manual segmentations, each time varying the position of the segmentations relative to

each other by translating one of the segmentations by a few pixels in random directions. Following each interpolation, the reconstructed segmentations are transformed back into the original slice plane by scaling the initial translation as a function of the averaging parameter t and applying it in the reverse direction. The result is a large set of possible segmentations for each unlabelled slice. The average of these segmentations is a probabilistic interpolation which takes into account the possible variability in slice plane orientation.

Alternatively, a computationally faster approach to model this uncertainty in BWA is to simply to apply a smoothing transformation that converts the BWA interpolation to a probabilistic weighted average. This is achieved by applying the error function to the interpolated distance function, $f_{A,B,t}(x)$, and scaling the values to lie in the range $[0, 1]$. The error function of a point x is given by

$$E(x) = erf\left(\frac{x}{\sigma}\right) = \frac{2}{\sqrt{\pi}} \int_0^{\frac{x}{\sigma}} e^{-t^2} dt. \quad (7)$$

2.3. Random forest for intensity-based classification

To perform intensity-based classification, we propose training a random forest (RF) classifier to distinguish the object of interest from background pixels [5, 6]. For a given dataset, a single classifier is trained using intensity information from the set of slices containing manual segmentations. The input to the classifier is the intensity images for these slices, and the corresponding multi-label images specifying foreground and background pixels. Pixels included in the manual segmentations are labelled as foreground, and a thin layer of pixels surrounding the boundary of each of the segmentations is labelled as background. These background pixels are identified by enlarging the boundary of the object by a few pixels using morphological dilation. During training, the intensities of voxels in a 5x5x5 patch form the features of the classifier. The trained classifier is then applied to the remaining unlabelled slices in the dataset, and assigns each voxel a probability of belonging to either the foreground or background.

3. EXPERIMENTS AND RESULTS

3.1. Dataset

The proposed method was evaluated on T2-weighted 9.4T *ex vivo* MRIs of the hippocampal region from 10 specimens, scanned at a 0.2 x 0.2 x 0.2 mm^3 isotropic resolution. In each dataset, on average 17 slices were selected, uniformly distributed along the length of the hippocampus, which extends over approximately 200 slices. On each of the selected slices, the hippocampus and parahippocampal gyrus were manually segmented using a single label to provide an outline of these structures to initialize building an unbiased population template of the medial temporal lobe [7]. Of the 17 manually segmented slices, every other slice was withheld for testing the algorithm. Therefore, interpolation was performed between only 9 slices, with an average spacing of 23 unlabelled slices between them, and evaluated on the 8 withheld slices. In all cases, manual segmentations for the first and last slice of the hippocampus was included in the interpolation (Fig. 2A).

3.2. Implementation

The algorithm for BWA was implemented using MATLAB and the RF implementation described in [8] was used. A combined interpolation was obtained by thresholding the voxel-wise average of the probability maps generated by each method. For evaluation, we computed the mean Dice Score Coefficient (DSC) between the interpolated segmentation and manually segmented ground truth on the withheld test slices. To evaluate the effectiveness of averaging multiple translated weighted averages to obtain a probabilistic BWA result, we applied translations of 10 pixels in 10 uniformly distributed directions. We found that applying smaller translations did not sufficiently capture variability in the slice plane, and varying the number of directions in which the translations were applied had no effect on the result. To evaluate the use of the error function to model BWA uncertainty, the error function was applied with a standard deviation of $\sigma = 5$, based on initial testing on a subset of datasets using different values of σ. Further evaluation on all the datasets shows that the performance of the proposed algorithm remains stable for $3 < \sigma < 5$, but drops off for larger values.

3.3. Comparative methods

We evaluated our proposed method against the Random walker (RW) algorithm [2] and morphology-based interpolation (MLB) [3]. The RW algorithm assigns each unlabelled pixel the probability that a ”random walker” starting at that pixel reaches one of the labelled pixels provided by the set of manual segmentations. MLB is an iterative object-based method that uses the conditional dilation operator to compute intermediate segmentations. The mean DSC was computed on the set of test slices for each of the datasets using each of the described interpolation methods. Additionally, we evaluated the reconstruction accuracy when using BWA on its own to assess the effectiveness of incorporating intensity information in the algorithm.

3.4. Results

Fig. 3 shows the mean DSC computed across all 10 datasets, using the different interpolation methods. When combining BWA with RF, simply applying the error function to the interpolated distance functions, $f_{A,B,t}(x)$ to generate probabilistic weighted averages, produces comparable results to the translation-based approach, and has the additional advantage of reducing computation time. Paired t-tests were used to

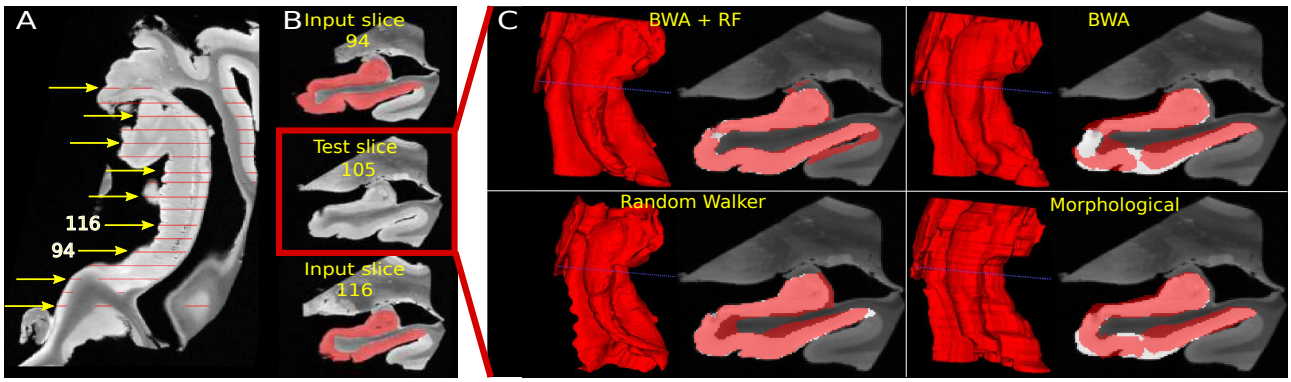

Fig. 2. (A) Example dataset with manually segmented slices shown in red. Input slices used to perform the interpolation are marked using yellow arrows. (B) Cross-sectional view of an unlabelled slice (slice 105) and the two nearest manual segmentations (slices 94 and 116) used for binary weighted averaging (C) Interpolated segmentations (red) obtained for test slice 105 using the four different methods are shown overlaid on the ground truth segmentation (white). For each method, a 3D reconstruction of the hippocampal region obtained by interpolating the entire set of input segmentations is also shown.

evaluate the performance of the proposed method of combining BWA and RF, at a single dataset level. Our method achieves a mean DSC of 0.89 ± 0.016 which is significantly higher than the mean DSC obtained using the RW algorithm and MLB. It also demonstrates lower variability in reconstruction accuracy. Fig. 2C shows the 3D reconstructions obtained using each of the interpolation methods, as well as the resulting segmentations on an example test slice. Using BWA on its own results in a 3D reconstruction with sharp turns in the boundaries between layers. Similarly, MLB results in visible steps between layers. Incorporating intensity information significantly improves reconstruction accuracy and produces smoother reconstructions compared to solely contour-based methods.

Even though a DSC of 0.89 might not be good enough for practical use, there is considerable anatomical variability in the direction of interpolation, and in practice, we would initialize the interpolation with more than 9 slices per 200 to account for this. Furthermore, the reported DSC values are based on measurements on test slices located halfway between two input manual segmentations. These slices likely have a lower segmentation accuracy than all the other interpolated slices since they are farthest from the input slices.

4. CONCLUSIONS AND DISCUSSION

We have proposed a novel interslice interpolation technique which uses both contour and intensity information from a subset of slices containing manual segmentations to annotate the remaining unlabelled slices in a dataset. It probabilistically combines a binary weighted averaging algorithm with random forest. Our experimental results are promising, and the proposed method shows significant improvement over the random walker algorithm and morphology-based interpolation. A version of the code for the proposed algorithm, implemented using the Insight Registration and Segmentation Toolkit (ITK) is available on GitHub[1]. One of the main challenges when interpolating segmentations of structures such as the hippocampus is the large shape-variability observed across cross-sectional slices. A future direction of this work is to develop a semi-automated segmentation tool which iteratively performs the interpolation, while allowing the user to interactively review the output at each step and correct/clean up the segmentations to refine the reconstruction.

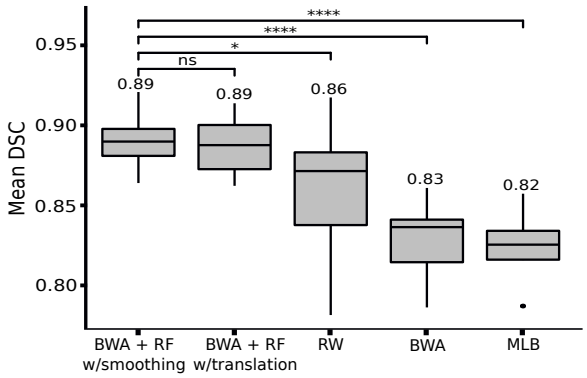

Fig. 3. Mean DSC obtained using the different interpolation methods (averaged across specimens). Asterisks indicate statistically significant differences in mean DSC between methods at a single dataset level (*: $P < 0.05$, ****: $P < 0.0001$)

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